

## Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

**Course:** Software Engineering Lab

**Project Name:** Academic Elective Bidding System (Key: AEB)

**Methodology:** Agile Scrum (Company-managed)

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Section: I

### 1. Jira Backlog & User Story Estimations

#### Epics and Functional Decomposition

- **Epic 1: Student Selection & Timetable Validation**
  - **Summary:** Epic 1: Student Selection & Timetable Validation
  - **Description:** Enable students to browse available elective offerings, allocate bidding credits across preferences, validate prerequisite eligibility, and detect schedule collisions.
- **Epic 2: Registrar Administration & Allocation Engine**
  - **Summary:** Epic 2: Registrar Administration & Allocation Engine
  - **Description:** Enable the Academic Registrar to set up course quotas and deadlines, execute the automated seat allocation algorithm, and publish final conflict-free timetables.

#### User Stories & Fibonacci Story Point Assignments

1. **Story 1.1: Course Catalog & Prerequisite Validation (AEB-3)**
  - **Format:** *As a Student, I want to browse available elective courses and have the system verify my prerequisites, so that I can select and bid only on courses I am eligible to take.*
  - **Priority:** High | **Story Points:** 3
2. **Story 1.2: Bidding Credit Allocation (AEB-4)**

- **Format:** *As a Student, I want to allocate 100 bidding credits across my ranked elective preferences, so that I can prioritize my chances of getting my most desired electives.*
- **Priority:** High | **Story Points:** 2

### 3. Story 1.3: Timetable Clash Detection (AEB-5)

- **Format:** *As a Student, I want to receive real-time alerts if my selected electives have overlapping lecture or lab slots, so that I can resolve timetable conflicts before final submission.*
- **Priority:** High | **Story Points:** 5

### 4. Story 2.1: Course & Quota Configuration (AEB-6)

- **Format:** *As an Academic Registrar, I want to configure offered electives, section seat caps, and bidding deadlines, so that the system enforces institutional constraints during the bidding window.*
- **Priority:** Medium | **Story Points:** 3

### 5. Story 2.2: Automated Allocation Solver (AEB-7)

- **Format:** *As an Academic Registrar, I want to run the automated batch allocation algorithm, so that seats are distributed based on bid weights and academic merit without exceeding section capacities.*
- **Priority:** High | **Story Points:** 8

### 6. Story 2.3: Publish & Distribute Timetables (AEB-8)

- **Format:** *As an Academic Registrar, I want to approve and publish final allocation reports and individualized schedules, so that registered students can download their conflict-free timetables.*
- **Priority:** High | **Story Points:** 3

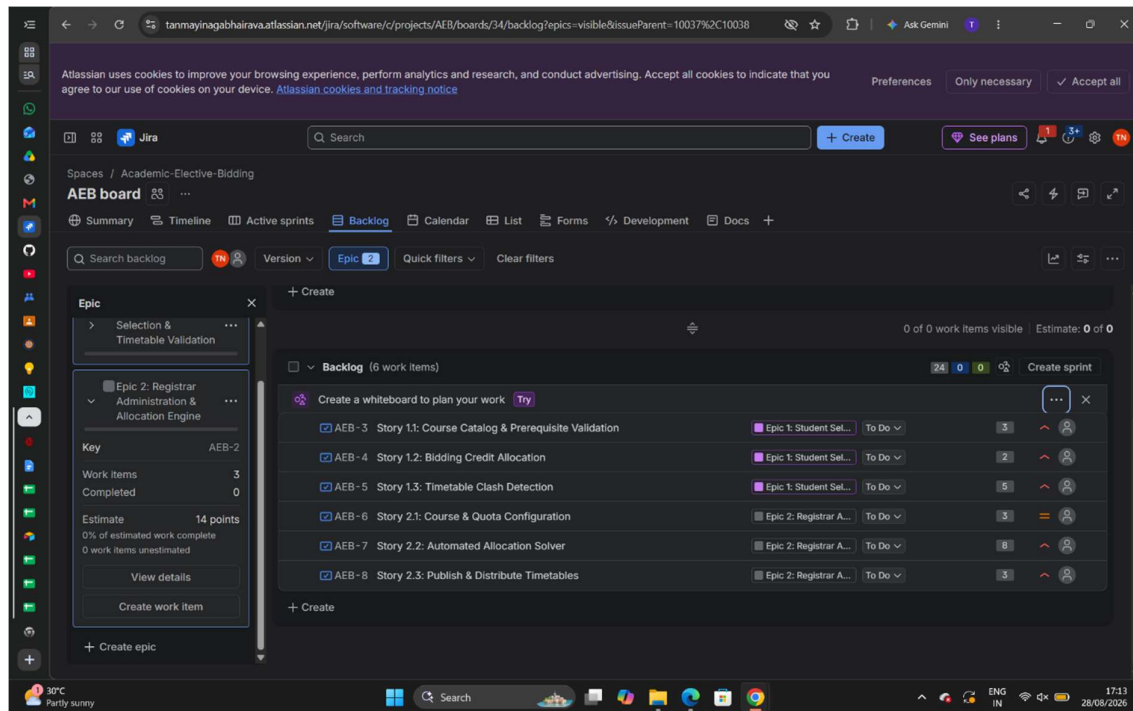


Figure 1: Jira Backlog showing Epics, prioritized User Stories, and assigned Story Points.

## 2. Sprint Execution & Board Simulation

The project was executed across two planned sprint cycles:

- **Sprint 1:** Focused on the student-facing bidding flow (Stories 1.1, 1.2, 1.3 totaling 10 story points).
- **Sprint 2:** Focused on administrative configuration, algorithmic allocation, and publication (Stories 2.1, 2.2, 2.3 totaling 14 story points).

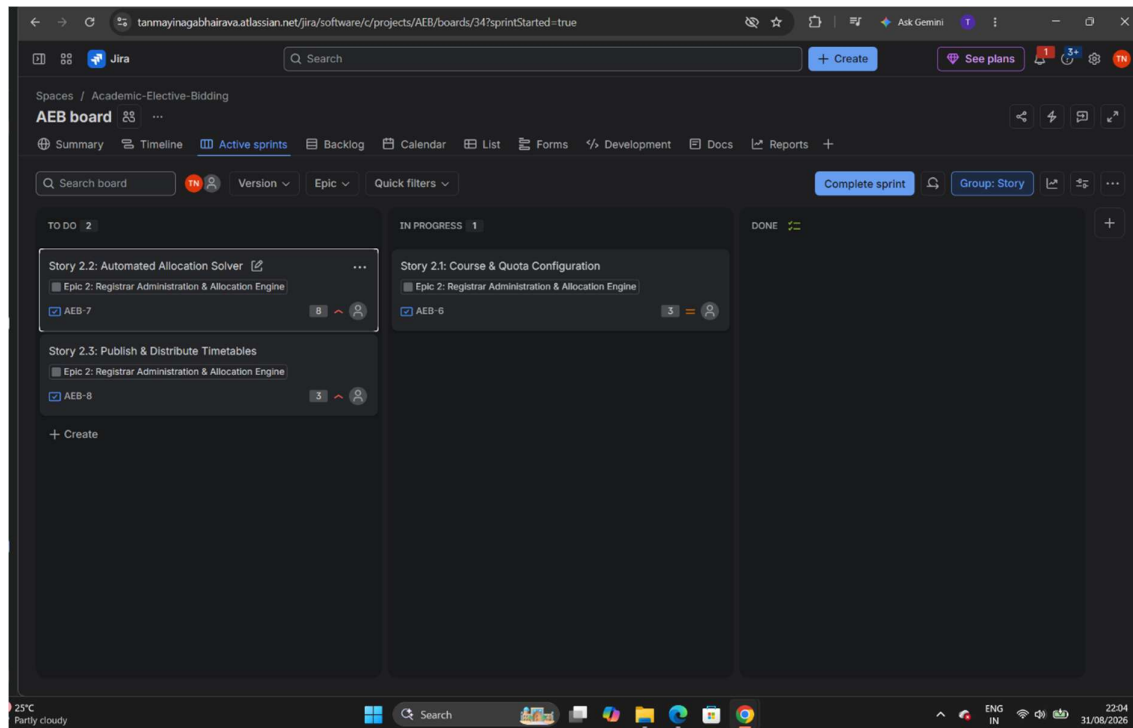


Figure 2: Active Sprint Board showing task progression through the Scrum workflow (To Do → In Progress → Done).

### 3. Burndown Chart Reports

#### Sprint 1 Burndown Analysis

Sprint 1 served as an in-lab simulation of end-to-end Scrum workflow mechanisms, burning down 10 committed story points upon completing initial prerequisite and validation tickets.

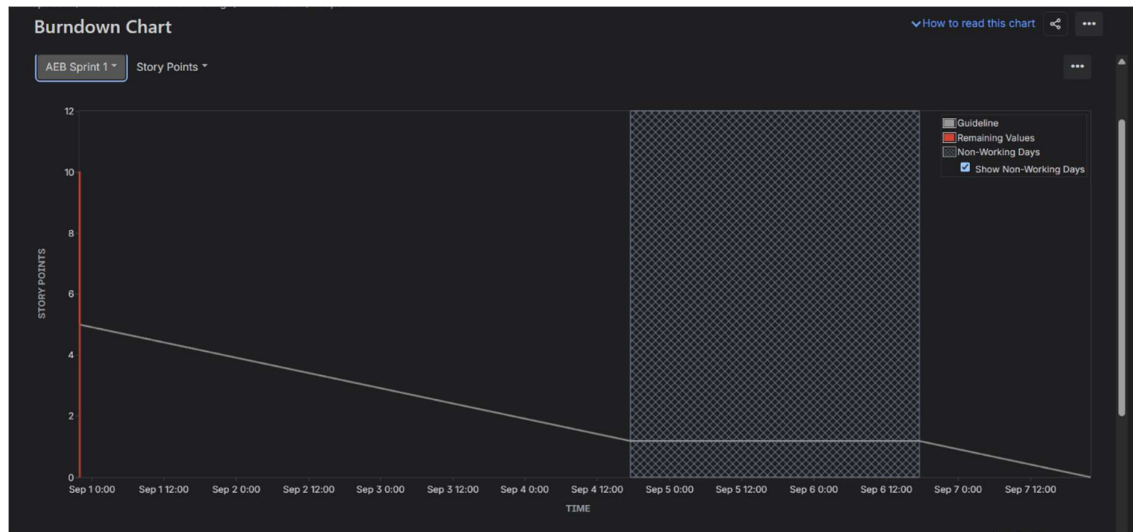


Figure 3: Jira Burndown Chart for AEB Sprint 1 ((Accelerated In Lab Simulation))

### Sprint 2 Burndown Analysis

Sprint 2 simulated a multi-day development cadence tracked between September 1 and September 4, burning down 14 points across discrete story completions:

- **September 1:** Sprint initiated with 14 committed story points.
- **September 2:** Story 2.1 completed, burning down 3 points (11 points remaining).
- **September 4 (Morning):** Story 2.2 (Solver engine) completed, burning down 8 points (3 points remaining).
- **September 4 (Afternoon):** Story 2.3 completed, bringing remaining story points to 0 ahead of the non-working buffer.



Figure 4: Jira Burndown Chart for AEB Sprint 2.

#### 4. Reflection Questions

##### 1. Did your estimations reflect the actual effort?

Yes, the non-linear Fibonacci scale (2, 3, 5, 8) did accurately reflect the relative complexity and uncertainty across the user stories. Input handling was necessary for simple form setups and credit entries (Story 1.2: 2 points; Story 2.1: 3 points). Conversely, algorithm tasks such as real-time timetable collision checking (Story 1.3 at 5 points) and the multi-variable automated batch allocation solver (Story 2.2 at 8 points) accurately reflected the higher architectural complexity, edge cases and testing effort.

##### 2. Was your backlog well-prioritized?

Yes, the backlog was purely prioritized on user value and functional dependencies. In Sprint 1, we planned to do prerequisite verification and capture student preferences to have valid data inputs. Sprint 2 was focused on administrative seat configurations, the allocation engine and schedule distribution, all of which require student bid data downstream.

##### 3. How did your simulated sprint align with your plan?

The sprint was executed very close to the planned timebox of 1 week. The 14 committed story points in Sprint 2 flowed smoothly from To Do, through In Progress, and into Done, over four active days, with no scope creep or unestimated spillover.

#### **4. What insights did the burndown chart give about your team's capacity?**

Comparing the two charts revealed basic Agile patterns. The vertical cliff of Sprint 1 illustrated how batched task movement hides granular progress, whereas Sprint 2 displayed a realistic stepped velocity trend. The sharp 8-point drop between September 3 and September 4 also revealed that big work items control team capacity. Breaking 8-point user stories into smaller 3- or 5-point tasks would result in an even smoother, predictable burn down against Jira's linear guideline.